Prediction Markets Explained

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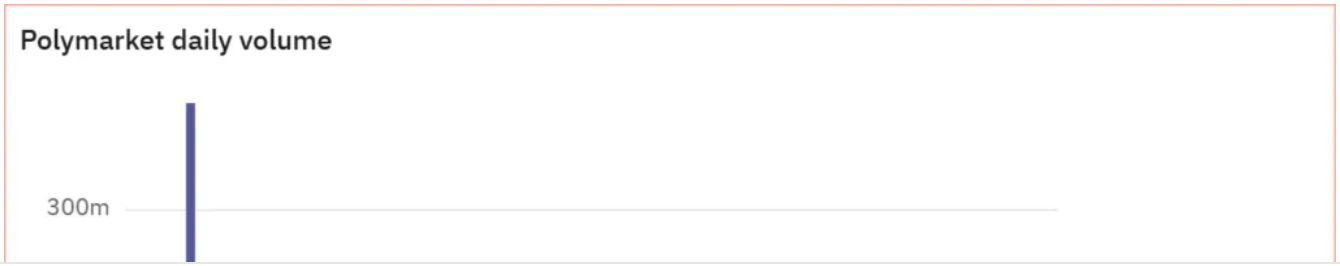
Why prices aren't probabilities, and what traders need to know.

Prediction markets had a breakout moment in 2024, with bets based on the US election attracting multi-billion-dollar volume (and huge media attention).

While things have calmed down since then, volumes are still far higher than pre-election averages. Prediction markets are here to stay.



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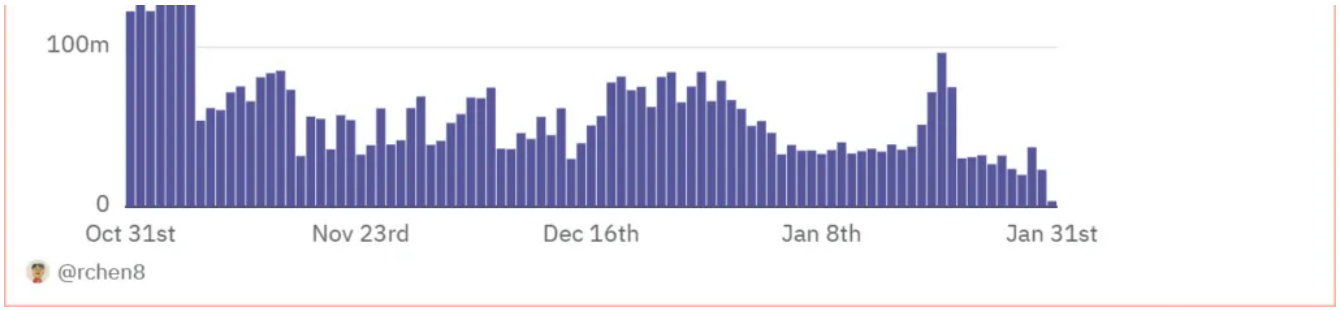


*Volume on **Polymarket** has faded since the election, but the market remains more liquid than before. Image: **rchen8 via Dune***

Prediction markets open up a whole new asset class, allowing investors to translate superior forecasts of real-world events directly into profits.

But there's still a tremendous amount of misinformation about these markets, with potentially serious implications for anyone trading event contracts.

Today, I'll cut through the noise and explain how prediction markets *really* work.

In the first half of this issue (free), I'll explain event contract mechanics and describe why prediction markets can be so valuable.

In the second half (paywalled), I dive into the finer details, exploring the two most important misconceptions of prediction markets:

1. Why event contract prices **aren't probabilities**
2. Why prediction markets **may not be an accurate forecasting tool**.

The second half gets technical — but don't be scared. I'll break things down simply, and if you're serious about trading in prediction markets, these are details you absolutely want to understand.

Let's go 👇

Note: All readers will learn something from this issue, but all-Access Pass holders will learn even more. **Grab an All-Access Pass** to get the full issue.

Let's go 👇

Brian Flaherty purchased his first mutual fund at 15. After graduating from UVA with a degree in Economics, he began advising institutions and high-net-worth investors as a strategist at a wealth management firm. Today, Brian helps investors uncover the best opportunities and make intelligent use of their capital. You can follow him on LinkedIn.

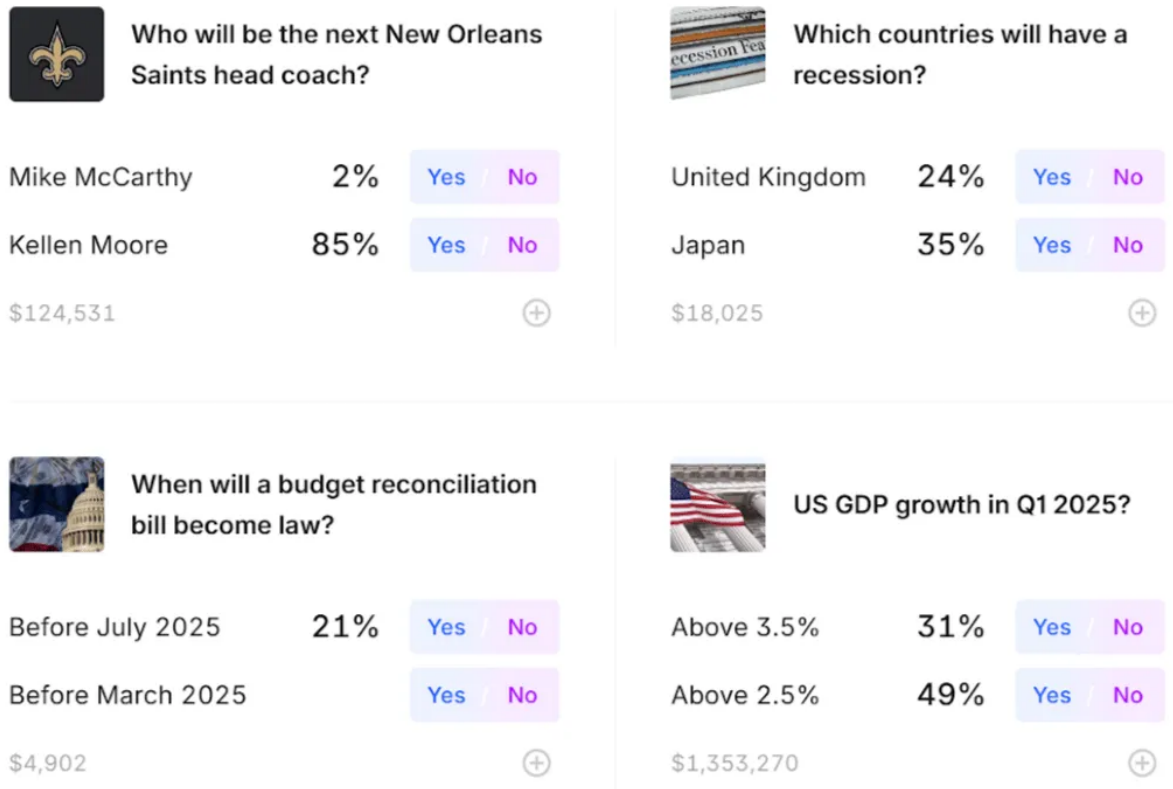
How do prediction markets work?

Prediction markets allow investors to trade a unique type of asset: **event contracts**.

Event contracts can be created on *almost* any topic. The most popular are politics, economics, sports, weather, and technology. (Some markets are off-limits, including death. Kalshi recently shut down a market **betting on the fate of Luigi Mangione**).

Event contracts have a **binary yes/no payoff** based on a specific future event. This is

what separates prediction markets from traditional betting platforms.



*A collection of **event contracts** on Kalshi. **Sign up here** to trade these markets.)*

See, in sportsbooks and casinos, you're betting directly against the *platform*. Event contracts, however, are a form of **peer-to-peer betting** in which you trade directly against *other bettors*.

(This is similar to Kutt, the P2P betting startup I recently wrote about).

Event contracts are traded on a market, and need standardized terms.

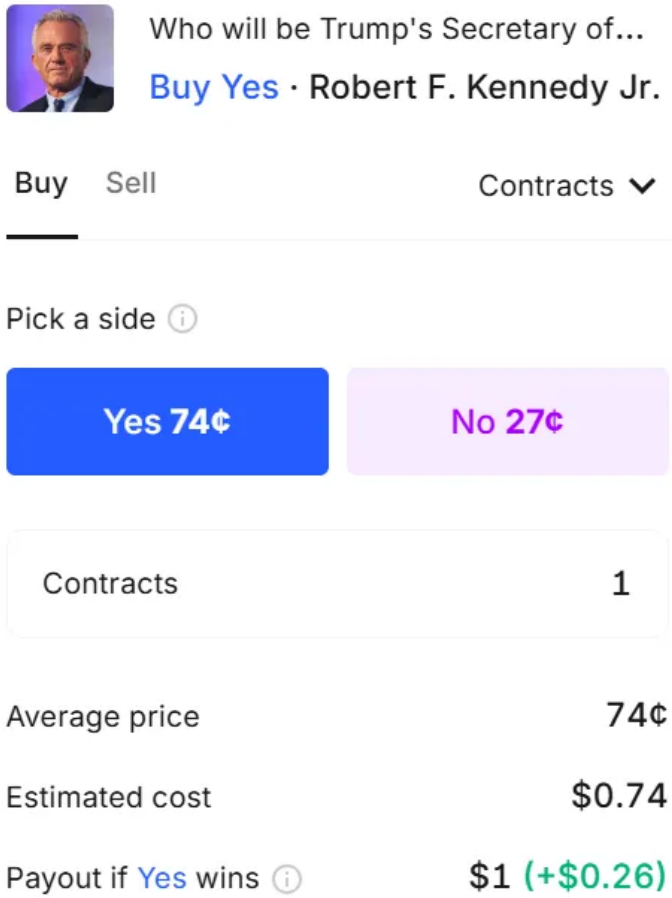
Here's how it works:

1. Each contract has two sides, *Yes* and *No*. You open a position by buying either *Yes* or *No*. There is only one winner.
2. The price of Yes and No **fluctuates** as new bettors place bets.
3. All contracts have a **$1** payout, which gets distributed entirely to the winning side.
4. Some events have multiple contracts, such as a presidential election with several candidates. But there is still only **one winner**.
5. You can sell your position **before or after** the contract ends.
6. You can only sell a position you own — **there is no shorting**.

(Technical note: Prediction markets don't actually have separate Yes or No contracts. "Under the hood," buying Yes is the same exact thing as selling No.)

For example: Suppose I wanted to bet that RFK Jr will be confirmed as Trump's Health Secretary.

- Right now, I could buy one Yes contract on Kalshi for about **74¢**. If I'm right, I'll get paid $1 when the contract settles, netting me a profit of 26¢ for a **35%** return.
- I could also bet **against** RFK's confirmation by buying a **No** contract for **27¢**. This is riskier, but if I'm right, I could earn a **170%** return (73¢ profit on 27¢ outlay).



Now, I must address a near-universal misconception that many people hold about prediction markets:

Prediction market prices do not precisely reflect the collective beliefs of an event occurring.

Yes, prediction market prices *approximately* reflect collective beliefs, and those approximations can be very close to realized outcomes.

But there are some technical reasons why prices stray from true probabilities, especially at the tails. You can already see evidence in the fact that the two RFK contract quotes add up to **$1.01** — which shouldn't be possible if prices reflect exact probabilities!

I'll explain this in more depth later on.

A short history of prediction market regulation

Before we go any further, we need to talk about the importance of CFTC, or the Commodity Futures Trading Commission.

CFTC regulation isn't just what sets apart legitimate derivatives exchanges from gambling platforms. It's what lets prediction markets operate in the first place.

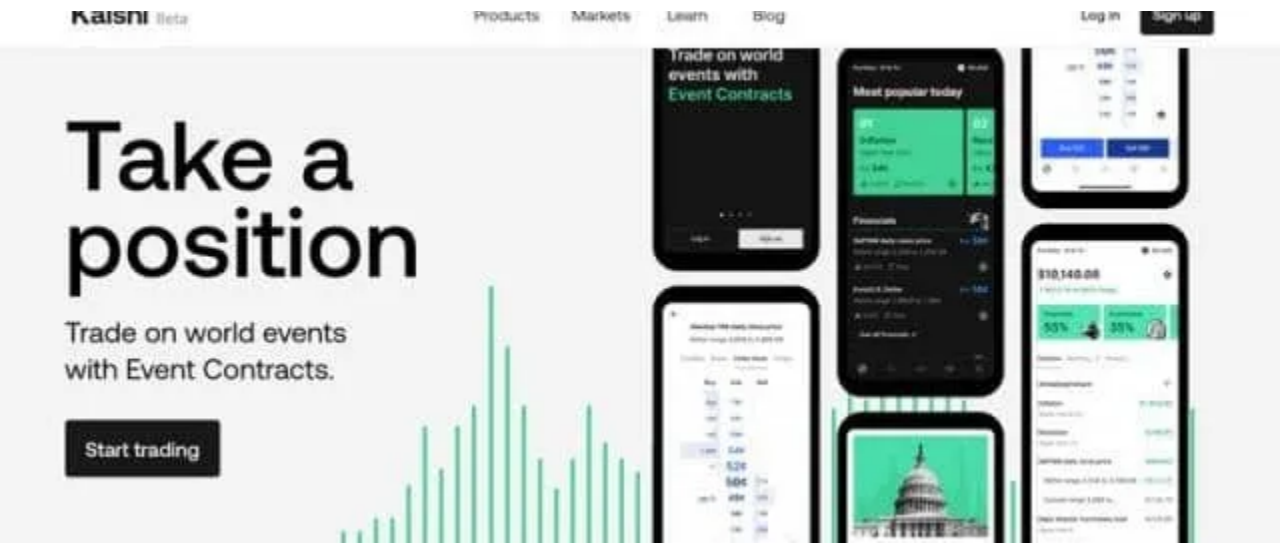
*The **CFTC** is the body that regulates prediction markets in America. They specialize in protecting the public from market manipulation and fraud. **Getting CFTC approval is a very big deal**.*

The past decade has *not* gone smoothly for several prediction market companies:

- In 2012, Irish prediction market **InTrade** was taken to court by the CFTC for violating various rules. Six months later, the platform was gone for good.
- In 2022, **Polymarket settled a $1.4m suit** with the CFTC for offering noncompliant event contracts.
- Seven months later, prediction market pioneers **PredictIt** also found themselves in hot water. The New Zealand-based platform was *banned from servicing US Investors* by the CFTC.

*Interestingly, the CFTC **did not publicly disclose** what PredictIt did to deserve the shutdown.*

Regardless, this is why CFTC regulation is so important. Without it, the agency has a clear path to probe and potentially penalize prediction markets. The big winner from all of this is **Kalshi**.



Kalshi is the first and only prediction market platform to get CFTC approval.

This is huge. Kalshi investors don't have to worry about the big bad agencies coming to shut them down, take away their investments, or smash their liquidity.

What's the purpose of prediction markets?

A more direct way to hedge

Prediction markets clearly offer value to people who just want to gamble on a wide range of events. But they can also provide value to savvy investors as a ***risk management tool***.

Let's say you believe China will invade Taiwan in the coming year. You could express

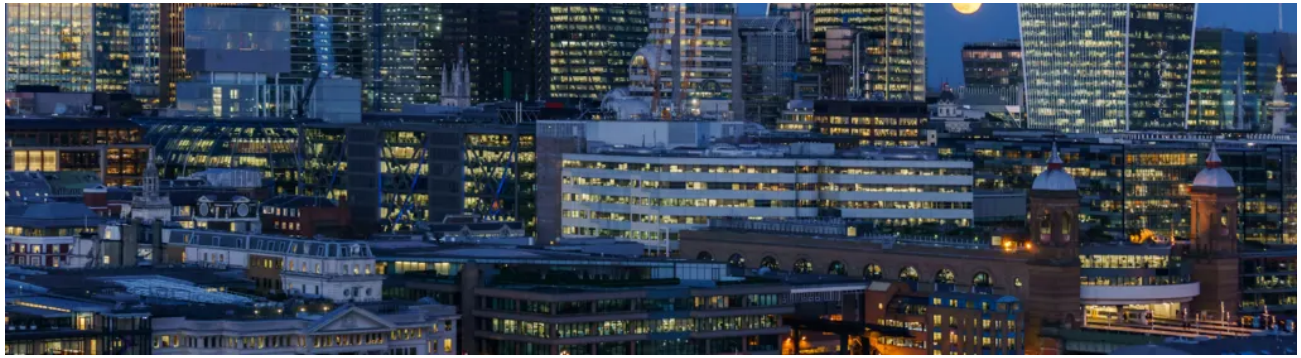
that view by going long American defense companies, or short TSMC.

But both of these positions are *imprecisely* related to the underlying event — they'll also be influenced by dozens of other factors.

In contrast, event contracts can offer a financial instrument **precisely tied** to the event you want to trade. This is a much cleaner way to speculate or hedge a specific outcome.

Back in 2022, we recorded a podcast with Tarek Mansour and Luana Lopes Lara, co-founders of Kalshi. As Tarek put it, this risk management idea was one of the key reasons behind Kalshi's creation:

> *When I was at Goldman in 2016, I was working at this exotics desk. And one of the main things we'd done that summer was structured products for institutions that wanted to get exposure to or hedge themselves against Brexit... It was a lot of different financial instruments. And the question was like, why is there no direct way of doing it?*



*You could have **bet on Brexit** through the pound, UK stocks, or gilts (UK government bonds). But an event contract would have offered a much **cleaner and simpler way to hedge**. Image of London by Colin*

But prediction markets aren't just valuable to individuals — they can benefit society.

Prediction markets can be more accurate than polling

Aggregating the insights of many people tends to be more accurate than relying on the prediction of just one individual, an idea known as the wisdom of the crowd. (This is the basic logic behind polls and surveys.)

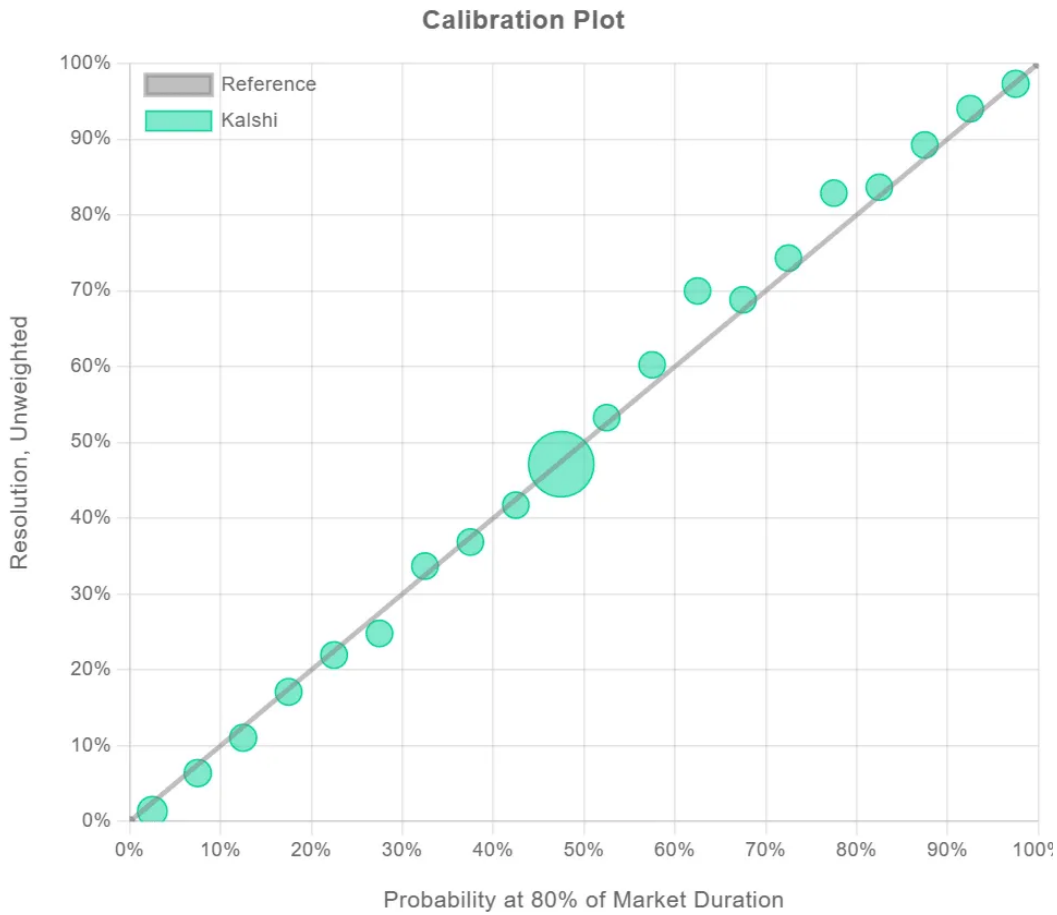
But prediction markets take this a step further by forcing forecasters to back up their beliefs with actual cash.

Prediction markets aren't perfect, but academic research indicates they can be more accurate than traditional polling in forecasting political events.

Calibration tests from Kalshi show that when aggregating over many events, prices

mostly align with the proportion of realized event outcomes.

(NOTE: Forecasting *individual* events is much different — in the paid section I'll explain why).



Interpreting prices as probability forecasts, this chart shows the market's forecast at 80% duration vs ***actual Kalshi outcomesi****. Source:* ***Calibration City***

Companies like Google and Ford have even deployed internal prediction markets to better forecast things like project timelines and expected sales.

That doesn't mean that prediction markets are without flaws, however.

Now let me show you why these markets can be **controversial and misinterpreted** — starting with the three reasons why prices aren't probabilities.

Why prices are *not* probabilities

It's tempting to treat prediction market prices as the "collective belief of the probability of the outcome."

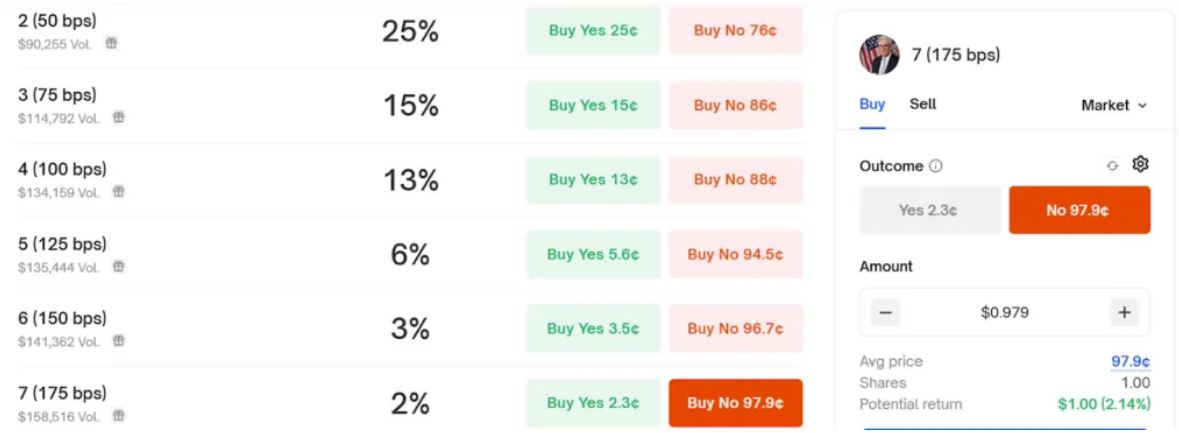
This idea is further entrenched because some markets even *quote* probabilities as if they were equivalent to prices.

But this isn't quite right. There are at least **three valid, rational reasons** that prediction market prices should systematically deviate from true underlying probabilities.

Reason 1: The risk-free rate

Prediction markets seem to exhibit a **persistent longshot bias**, with things that very clearly aren't going to happen nonetheless trading at positive prices.

For example, let's look at the Kalshi market for the **number of Fed rate cuts in 2025**. Specifically, observe that markets seem to expect a **2% chance** of *seven* Fed rate cuts this year (?!)



Now, I'm a smart guy, and I feel *very* confident that the Fed won't cut rates 7 times this year. Heck, there are only 7 more scheduled Fed meetings for the entire year, and we'd have to see truly rapid economic deterioration for cuts to occur at every single meeting!

However, that doesn't necessarily mean I should buy the No contract! Here's why:

- For every contract I buy at 97.9¢, I could earn 2.1¢ in profit for a return of **2.15%**.
- This contract doesn't settle until the end of the year. If I bought those contracts on February 1st, my annualized return would be roughly **2.34%**.

- Why would I ever accept a risky return of 2.34% when I can currently earn **4.17%** risk-free on a 1-Year Treasury?

Basically, because of the risk-free rate, it's not always worth it for investors to take the other side of longshot trades. That can result in persistent price distortions that make improbable outcomes appear *more likely than they truly are*.

This also helps explain why long-shot contracts tend to fall to 0 as the end of the contract approaches: arbitraging small price-probability discrepancies becomes more attractive on a time-value basis.

The risk-free rate has a fairly straightforward impact on contract prices, but it's actually a subset of a more complex issue: the existence of **alternative investment opportunities**.

Reason 2: Opportunity cost

Let's continue with our Fed rate cut example.

Suppose that you are absolutely *certain* that the Fed will cut 7 times this year — there's just no doubt in your mind.

Does that mean you should buy the Yes contracts? Possibly, but it's not a foregone conclusion.

Assuming that you have a limited amount of capital, you need to find the most **capital-efficient** way to express your view. And there are *plenty* of other asymmetric ways to bet on interest rates falling, including:

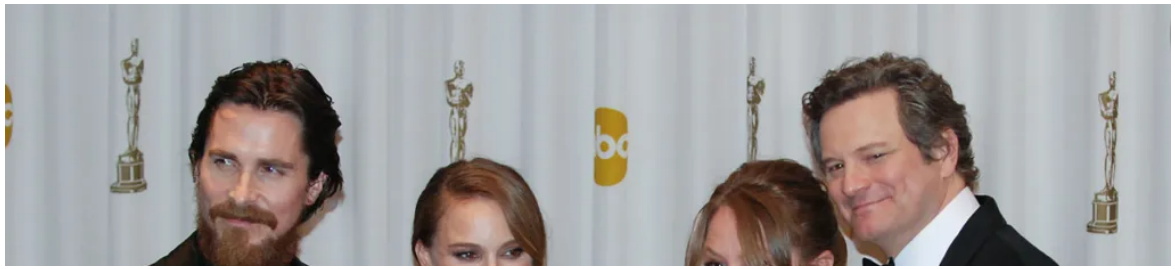
- Going long high-duration **Treasury bonds on margin**
- Buying deep, out-of-the-money **Federal Funds options**

This problem is even more severe if you consider *liquidity constraints* in prediction markets.

The **Kalshi order book** (**sign up to view it**) shows me that there are only about 4,000 contracts available for sale at less than 4¢. If I want to do this trade in serious size while preserving my asymmetry, I'll probably need to look to larger markets.

The bottom line is that even a bettor with perfect information isn't always incentivized to arbitrage price-probability differentials.

Traders need to find the most capital-efficient way to express their views, which *may* live on prediction markets, but they may live elsewhere.



*One odd result of this argument is that markets based on **pop culture** (like the Oscars) have more accurate probabilities than markets based on business or economics since there are fewer ways to trade on those forecasts. Image: Carlos Lazo*

Reason 3: Bid-ask spread

Earlier, I mentioned that the RFK contracts didn't sum to $1. The reason is due to the **bid-ask spread**.

All financial exchanges have a bid-ask spread, and prediction markets are no different. The bid-ask spread represents the cost of *immediacy* (aka the fee that liquidity providers earn).

If I wanted to take a round-trip in the RFK market at quoted prices, I would need to buy the Yes at a 74¢ ask and then immediately sell it at a 73¢ bid.

Remember that selling the Yes is equivalent to buying the No. And indeed, selling Yes at 73¢ implies buying No at 27¢.

In other words, the price-implied probabilities at the same instant are 74% and 27%, summing to 101%. **This extra 1% reflects the distorting impact of the bid-ask spread.**

This effect can be more noticeable in markets with low liquidity. The current market on Kalshi for whether the stranded astronauts will be home by July 1st somehow implies both a **98%** chance that they will return by that date, and also a **7%** chance that they won't — clearly violating basic probability.

Are prediction market prices actually informative?

In the previous section, I argued that prediction market prices can only *approximate* underlying probabilities.

Now, I'm going to explore a more **controversial idea…**

Contrary to popular opinion, careful data analysis shows that prediction market prices may not actually offer much useful information for *forecasting* individual events.

To understand why, we'll need a brief statistics refresher.

(This next part will get a bit dense, but if you're truly interested in these markets, it's absolutely worth your time.)

Calibration vs accuracy

Earlier, I showed a chart showing that Kalshi prices appear well-calibrated to actual event outcomes.

However, **calibration** is not the same thing as **accuracy**.

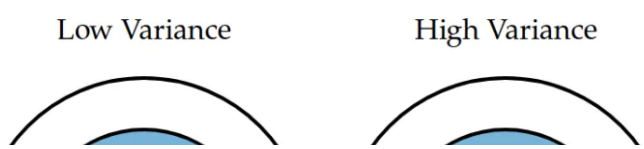
To understand the difference, imagine if I were trying to build a predictive model to forecast the chance of rain in a wet region over the course of a year.

Based on historical data, I see that it tends to rain about half of all days. Therefore, I develop a simple model — every day, I predict a 50% chance of rain.

Over the year, it ends up raining exactly 183 days while being dry the other 182. In the aggregate, my model is extraordinarily **well-calibrated** — events that I predicted would occur 50% of the time ended up happening about 50% of the time.

But on a day-to-day basis, my model is *absolutely useless*!

Flipping a coin to determine whether it will rain each day offers no predictive value. In statistical terms, our model has very **poor accuracy**.



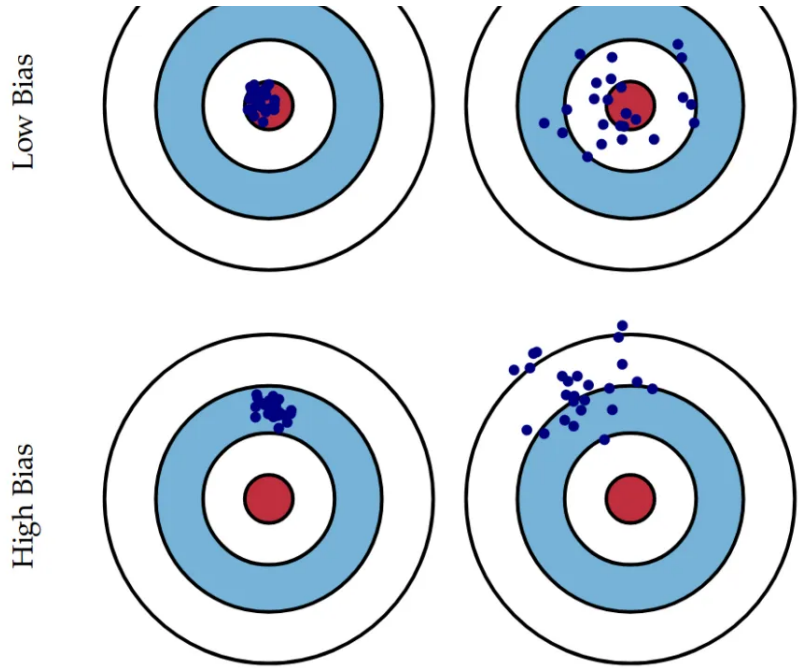


Fig. 1 Graphical illustration of bias and variance.

The calibration/accuracy distinction has a lot of conceptual overlap with ***bias/variance in machine learning****. Image:* ***Scott Formtann-Roe***

So, how do you measure the accuracy of a predictive model?

One of the best ways is with a **Brier Score.**

What is the Brier Score?

The Brier Score can be used to determine whether prediction market

prices offer useful forecasting information.

You don't need to understand all the intricacies. I'll give you the main idea.

A Brier score attempts to measure the **average error** a model has in forecasting individual events (technically, the average *squared* error).

Here's how it works:

1. Events that occur are assigned a value of 1, and events that don't occur get a value of 0.
2. Suppose our model predicts the likelihood of an event at .90 and

🔒This section is locked!

That's the end of what we can give away for free. To read the rest you'll need the All-Access Pass. 🎟️

Psst — If you want to make money in prediction markets, get the pass.

It also lets you ask Brian questions directly in a private space in the **Alts Community**.

This quote from All-Access Pass holder John Nikolaou says it all:

> *"Thanks again for your DXYZ feature!* ***I invested, and paid for my entire Spain vacation****, as it cruised to the mid $90s from going public around $8-10/share. That's the kind of tax problem I like to have."* – John Nikolaou

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That's it for today!

Come **chat with me about this issue** in the Alts community!

See you there,
Brian

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- This issue was sponsored by **GrowthSchool** and **Kalshi**
- Kalshi had zero editorial input or oversight in this issue.
- After researching this issue, Brian may place some trades in prediction markets. He held no positions in prediction markets at the time of writing or publication.

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